Charged Lepton Flavor Violation

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Abstract

We review how the chiral nature of fermions in the Standard Model with massless neutrinos leads to a mixing of quark flavors but not to the mixing of charged lepton flavors. A standard argument behind the existence of Cabibo-Kobayashi-Maskwa (CKM) matrix is given.

1 Introduction

The Standard Model of particle physics is one of the most well-tested theories in all of sciences. This does not mean, however, that it describes the complete story of the universe. For starters, it does not describe the effects of the force of gravity at short distances. It also does not explain the asymmetry in the number of particles and anti-particles in the universe. For a recent discussion of more questions that the Standard Model does not answer, see for example, [1].

The detection of a charged lepton flavor violating transition would open one avenue into the study of physics beyond the Standard Model [2]. As we review below, such transitions are not allowed in the Standard Model when neutrino masses are absent. The arguments presented here are standard, and can be found in most textbooks and lecture notes on Particle Physics. The reader may consult, for example, chapter 20 of [3] for background or more details on the subject.

2 Flavor violation in SM with massless neutrinos

We collect here some basic facts about the Standard Model of physics. In particular, we focus on the electroweak sector; the color degrees of freedom of quarks and gluons will play no role in the following discussion. The gauge group describing this force is the product group $SU(2) \times U(1)$. In the following we denote a representations of this group as $\boldsymbol{n}_q$ where $\boldsymbol{n}$ denotes the n-dimensional representation of $SU(2)$ and q denotes the $U(1)$ charge, often also known as hypercharge. The standard model contains a Higgs field, which transforms in $\mathbf{2}_{1/2}$. As it obtains a vacuum expectation value, which in a particular gauge, may be written as

$$\langle \Psi \rangle = \begin{pmatrix} 0 \\ \frac{v}{\sqrt{2}} \end{pmatrix} \tag{2.1}$$

the gauge group $SU(2) \times U(1)$ breaks down to $U(1)_{\text{em}}$. This group is generated by $T_3 + Q$ where T_3 is the generator of Cartan of $SU(2)$ and Q is the generator of $U(1)$.

The standard model is a chiral theory, i.e. the left-handed and right-handed fermions transform differently under the action of $SU(2) \times U(1)$. In particular, the left-handed fermions transform in three copies of $\mathbf{2}_{-1/2} \oplus \mathbf{2}_{1/6}$ while the right-handed fermions transform in three copies of $2(\mathbf{1}_{-1}) \oplus \mathbf{1}_{2/3} \oplus \mathbf{1}_{-1/3}$.

Field	$SU(2)$	$U(1)$	$U(1)_{\text{em}}$
Higgs field Ψ	$\mathbf{2}$	$+\frac{1}{2}$	0
A right-handed u-type quark $(u_R)_i$	$\mathbf{1}$	$+\frac{2}{3}$	$+\frac{2}{3}$
A right-handed d-type quark $(d_R)_i$	$\mathbf{1}$	$-\frac{1}{3}$	$-\frac{1}{3}$
A left-handed doublet of quarks $(\phi_L)_i$	$\mathbf{2}$	$+\frac{1}{6}$	$\begin{pmatrix} +\frac{2}{3} \\ -\frac{1}{3} \end{pmatrix}$
A left-handed doublet of leptons $(\psi_L)_i$	$\mathbf{2}$	$-\frac{1}{2}$	$\begin{pmatrix} 0 \\ -1 \end{pmatrix}$
A right-handed electrically charged lepton $(\psi_R)_i$	$\mathbf{1}$	-1	0

Table 1: The $SU(2) \times U(1)$ representation content of the Standard Model

Since chiral theories do not admit mass terms, the only way to give masses to fermions is through their coupling with Higgs field. In particular, the most general Yukawa terms with this field content are of the following form.

$$\mathcal{L}_{\text{Yukawa}} = -(\psi_L)_i^\dagger (Y_L)_{ij} \Psi (\psi_R)_j - (Y_d)_{ij} (\phi_L)_i^\dagger \Psi (d_R)_j - (\phi_L)_i^\dagger (Y_u)_{ij} i\sigma_2 \Psi (u_R)_j + c.c. \quad (2.2)$$

Here Y_k, for $k = u, d, L$ are 3×3 matrices of Yukawa couplings; the rest of the symbols are explained in Table 1. When the Higgs field acquires vacuum expectation value (2.1), the product of $\langle \Psi \rangle$ and Y_k get the interpretation of mass matrices for the fermionic fields. In this case, the eigenstates of the mass matrices get interpretation of quarks with the corresponding eigenvalues being their respective masses. In order to diagonalize the matrices Y_k, we make unitary transformations of the fields such that the transformed matrix is diagonal and has real entries. For example, for the first term in equation (2.2), if we let

$$(\psi_L)_i \to (\tilde{\psi}_L)_i = L_{ij} (\psi_L)_j \quad ; \quad (\psi_R)_i \to (\tilde{\psi}_R)_i = E_{ij} (\psi_R)_j \quad (2.3)$$

then the Yukawa matrix in the new basis $\tilde{Y}_L = L Y_L E^\dagger$ has this property for some unitary matrices E and L. As $(\tilde{Y}_L)_{ij} = \delta_{ij} m_i$ in this basis, the lepton Yukawa interaction takes the following form which is appropriate for pairing left-handed and right-handed leptons into a Dirac fermion.

$$\mathcal{L}_{\text{Lepton,Yukawa}} = \sum_i (\tilde{\psi}_L^\dagger)_i m_i (\tilde{\psi}_R)_i + c.c. \quad (2.4)$$

Things are not so straightforward for the quark Yukawa terms. In particular, since the left-handed quarks make appearance in both the second and the third term of equation

(2.2), there is a possibility that the unitary transformations acting on ϕ_L that diagonalize Y_d and Y_u do not coincide. Focusing on the term involving d-type quarks, we learn that for some unitary matrices Q and $D^\dagger$, the matrix $\tilde{Y}_d = QY_dD^\dagger$ is diagonal, giving the second component of $\tilde{\phi}_L = Q\phi_L$ the interpretation of left-handed d-type quarks. Naively, their $SU(2)$ counterparts, i.e. the first components of $\tilde{\phi}_L$, are the left-handed up-type quarks, however this fails as these states do not diagonalize Y_u.

In order to diagonalize Y_u, we transform it to a new basis and write $\tilde{Y}_u = Q'Y_uD^\dagger$, giving the top components of $\tilde{\phi}'_L = Q'\phi_L$ the interpretation of left-handed u-type quarks. The discrepancy between $\tilde{\phi}_L$ and $\tilde{\phi}'_L$ is captured as

$$\tilde{\phi}_L = Q(Q')^\dagger\tilde{\phi}'_L \tag{2.5}$$

The unitary matrix $V \equiv Q(Q')^\dagger$ is called the Cabibo-Kobayashi-Maskawa (CKM) matrix and measures the magnitude of quark flavor mixing. The experimentally-measured values of its entries are given below [4].

$$V = \begin{pmatrix} 0.97446 \pm 0.00010 & 0.22452 \pm 0.00044 & 0.00365 \pm 0.00012 \\ 0.22438 \pm 0.00044 & 0.97359 \pm 0.00010 & 0.04214 \pm 0.00076 \\ 0.00896 \pm 0.00024 & 0.04133 \pm 0.00074 & 0.999105 \pm 0.000032 \end{pmatrix} \tag{2.6}$$

3 Conclusion

In this note, we reviewed the standard argument behind the existence of quark flavor mixing in the Standard Model. As explained in [2], there are several different ways in which one could modify the Standard Model in a consistent way in order to include the possibility of charged lepton flavor violation. The author believes that it would be nice for him to understand these procedures in the future.

Acknowledgements

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Charge-Parity Symmetry Violation

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Abstract

We review the theoretical background and experimental results related to the Charge-Parity (CP) symmetry violation in electroweak theory. Its origin in the complex nature of CKM matrix is discussed, and related to the recent discovery of CP violation in D^0-meson decays.

1 Introduction

It is well known amongst the practitioners of physics that symmetries play a crucial role in building an understanding of our universe. It is appreciated less however, that a violation of an expected symmetry can provide crucial insights into the phenomenology. One example of such a violation is the charge-parity (CP) symmetry of electroweak interactions. This symmetry exchanges a particle with the mirror image of its anti-particle. A violation of this symmetry implies that these two species interact differently in the decays and scattering processes. The purpose of this essay is to review some theoretical and experimental results in the understanding of this violation.

In section 2, we review the theoretical aspects of the subject. In particular, this involves the study of the complex nature of Cabibo-Kobayashi-Masakawa (CKM) matrix. This is followed by a discussion of old and some new experimental results in section 3.

2 Theoretical Background

The electroweak sector of the Standard Model has a Lagrangian description in terms of an $SU(2) \times U(1)$ gauge theory. The details of this gauge theory were reviewed in [], where it was also explained that the Yukawa interactions in this model take the following form.

$$\mathcal{L}_{\text{Yukawa}} = -(\psi_L)_i^\dagger (Y_L)_{ij} \Psi (\psi_R)_j - (Y_d)_{ij} (\phi_L)_i^\dagger \Psi (d_R)_j - (\phi_L)_i^\dagger (Y_u)_{ij} i\sigma_2 \Psi (u_R)_j + c.c. \quad (2.1)$$

We had also reviewed that the diagonalization of mass-matrices for up-type and down-type quarks is carried out in different bases, leading to an interaction between these quarks. The amplitudes of this interaction are encoded in a 3×3 unitary matrix V_{CKM},

commonly known as Cabibo-Kobayashi-Masakawa (CKM) matrix. Here we are interested in studying the action of charge-parity (CP) symmetry on the Yukawa terms in the Lagrangian. This action is given as follows.

$$CP : (\phi_L)_i^\dagger \Psi (\phi_R)_j \to (\phi_R)_j^\dagger \Psi^\dagger (\phi_L)_i \tag{2.2}$$

In order for the Lagrangian to be invariant, the Yukawa coupling matrices Y_k must be real, i.e. $Y_k^\star = Y_k$. This implies in return that the CKM matrix defined in [1] must be real, i.e. $V_{CKM}^\star = V_{CKM}$. However, experimental observations show that this matrix is in fact complex, implying that the CP symmetry is violated in electroweak interactions.

To understand the complex nature of the CKM matrix, we note that a unitary 3×3 matrix has nine independent real parameters. Five out of these parameters, however, can be absorbed into the definition of quark fields, leaving with four physical parameters. A standard paramerization of CKM matrix uses a phase δ and three Euler angles θ_{ij} where $i, j \in \{1, 2, 3\}$ and $i < j$. In terms of these parameters the CKM matrix is written as

$$V_{CKM} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix} \tag{2.3}$$

where $s_{ij} = \sin\theta_{ij}$ and $c_{ij} = \cos\theta_{ij}$. This parameterization of the CKM matrix is also adopted by the Particle Data Group [2]. Clearly the measurement of CP violation should not depend on a particular parameterization. For this purpose, an invariant J known as *Jalskog invariant* is defined that measures CP violation. In the parametrization given above, it takes the form

$$J = c_{12}c_{13}^2 c_{23} s_{12} s_{13} s_{23} \sin\delta \tag{2.4}$$

If any of the parameters c_{ij}, s_{ij} or δ are zero, there will be no CP violation.

3 Experimental Results

The violation of CP symmetry may be observed in experiments involving neutral meson decays. In particular, say P^0 is a neutral meson; standard examples are the neutral B-meson $B^0(d\bar{b})$, the neutral Kaon $K^0(d\bar{s})$ and the neutral D-meson $D^0(c\bar{u})$. The action of CP symmetry replaces it with its anti-particle $\overline{P}^0$. There exists a violation in this symmetry if the decay rate of P^0 into a final state f differs from that of the decay rate of $\overline{P}^0$ into CP-conjugated final state $\bar{f}$. In this case, one may define a measure of CP-asymmetry as

$$A_{CP}(f, t) := \frac{\Gamma(P^0(t) \to f) - \Gamma(\overline{P}^0(t) \to \bar{f})}{\Gamma(P^0(t) \to f) - \Gamma(\overline{P}^0(t) \to \bar{f})} \tag{3.1}$$

For example, the experimental value of this number for $B^0 \to \pi^- + K^+$ decay is -0.098 ± 0.012 [3]. A recent paper has reported the violation of CP symmetry in the

decays of D^0 meson [4]; this being the first observation of CP violation in the charm sector. In particular, they study two different decay channels of D^0 observed in the 2011-2018 run of LHCb — $D^0 \to K^- + K^+$ and $D^0 \to \pi^- + \pi^+$, and compare the time-integrated CP asymmetry.

$$\Delta A_{CP} := A_{CP}(K^- + K^+) - A_{CP}(\pi^- + \pi^+) \tag{3.2}$$

This number takes a non-zero value. In particular,

$$\Delta A_{CP} = (-15.4 \pm 2.9) \times 10^{-4} \tag{3.3}$$

measured with the statistical significance of 5.3 standard deviations.

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Dark Matter and Cross Sections

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October 24, 2019

Abstract

After presenting a brief review of the evidence of the existence of dark matter in the context of FRW cosmology, we discuss the validity of weakly interacting massive particles (WIMPs) as candidates for dark matter on both theoretical and experimental grounds.

1 Introduction

Dark matter is a non-baryonic form of matter that constitutes approximately 85% of the matter in the universe. The evidence for its presence has come from astrophysical observations; though it is largely believed that its properties will be explained by beyond the Standard Model particle physics. In this essay, we briefly review first its origins in FRW cosmology and observations of cosmic microwave background (CMB) (c.f. sec. 2.1). Then we discuss a possible theoretical explanation of the astrophysical observations through a model of dark matter as weakly interacting massive particles (WIMPs) (c.f. 2.2). Lastly, we discuss recent experimental results in relation to this model (c.f. sec. 3).

We remark here that the contents of section 2 are largely based on the lecture notes [1] and the contents of section 3 are based on [2] and [3].

2 Theoretical Background

2.1 What is Dark Matter?

In the Friedmann-Robertson-Walker cosmology, the expansion of our universe is controlled by a scale parameter $a(t)$ that appears in the metric as a coefficient of the spatial distances: $ds^2 = -dt^2 + a^2(t)d^2x^3$. A parameter, known as Hubble constant then defines the rate at which the universe may expand or contract. It is defined as $H = \frac{1}{a(t)}\frac{da}{dt}$. The word constant in Hubble constant is a misnomer as H is in fact a function of time. At the current stage of our universe, experimental results dictate that H takes the following value.

$$H \approx 70\text{km}/(\text{Mpc.s}) = 1.5 \times 10^{-33}\text{eV} \tag{2.1}$$

A dimensionless quantity, often denoted h is defined in terms of the Hubble constant as

$$h := \frac{H}{100\text{km}/(\text{Mpc.s})} \tag{2.2}$$

At the current stage of our universe, $h \approx 0.7$. Another cosmological quantity of interest is the matter content of the universe. A natural way to quantify it is to use the mass density ρ_m, however a dimensionless quantity denoted Ω_m is often used instead. Ω_m is defined in terms of ρ_m as $\Omega_m = \frac{\rho_m}{\rho_{\text{crit}}}$ where ρ_{crit} is the critical density separating an expanding universe from a collapsing universe with the cosmological constant Λ being zero.

The matter content of the universe comes from two contributions: the baryonic matter density Ω_b and the dark matter density Ω_χ. The experimental value of Ω_χ is given through the measurements of the anisotropy of the cosmic wave background (CMB) and the spatial distribution of galaxies.

$$\Omega_\chi h^2 = 0.1186 \pm 0.0020 \tag{2.3}$$

Since the subject of general relativity has passed several experimental tests with flying colors, it is believed that the presence of dark matter can be explained through beyond the Standard Model particle-physics models. To this end, there are several proposed candidates for dark matter in terms of non-baryonic particles. Here we discuss one of them: the weakly interacting massive particles, alternatively called WIMPs.

2.2 What are WIMPs?

In particle physics, a WIMP is any particle of masses larger than a few GeVs that interacts with other particles with a strength comparable to that of the weak force. Assuming that the mass of such a particle χ is much less than the mass of Z-boson, the cross section of its annihilation process into a fermion-anti-fermion pair ($\chi + \chi \rightarrow f + \bar{f}$) mediated through an exchange of a W-boson is

$$\sigma = \frac{\pi \alpha^4 m_\chi^2}{c_w^4 m_Z^4} \tag{2.4}$$

where c_w is related to the weak mixing angle θ_w as $c_w = cos^2\theta_w$. Assuming that a WIMP is in thermal equilibrium with particles of the Standard Model at a high temperature, the number density of χ at lower temperaturess is effectively constant compared to the expansion rate of the universe dictated by the Hubble constant H. One can then compute the dark matter density ρ_χ at the current time of the universe T_0 through thermodynamical calculations.

$$\rho_\chi(T_0) = 3 \times 10^3 \frac{m_Z^4}{m_\chi^2 M_{\text{pl}}^2} T_0^3 \tag{2.5}$$

It gives the following expression for dimensionless dark matter density.

$$\Omega_\chi h^2 = \frac{\rho_\chi h^2}{\rho_{\text{crit}}} \approx 0.1 \frac{\text{pb.c}}{\langle \sigma v \rangle} \approx 0.12 \left(\frac{13\ \text{GeV}}{m_\chi} \right)^2 \tag{2.6}$$

Here v is the relative velocity between the two WIMPs written in the center of mass frame and $\langle ... \rangle$ is the notation of taking thermal averaging under Boltzmann distribution. This result is equivalent to the one given in equation(2.3) for certain values of m_χ. This observation is known as the "WIMP miracle". It brings forth the following question: can WIMPs be detected experimentally through direct or indirect methods. This question is the subject of next section.

3 Experimental Results - DAMA/LIBRA vs other experiments

The experimental strategy for the detection of WIMPs is based on measuring a recoil of a nucleus after scattering with these particles. For this purpose, the DAMA/LIBRA experiment uses thallium-doped sodium iodide (NaI(TI)) scintillation detectors. Experimental results show the possibility of WIMPs of mass close to 50 GeV scattering off of iodide with cross section $\sigma = 7 \times 10^{-6}$pb or of mass close to $\sim$ 10 GeV scattering off of sodium with cross section $\sigma = 2 \times 10^{-4}$pb. This experiment released results for 14 consecutive years, with an annual modulation of the signal in 2-5 keVee bin and at a significance of more than 5σ.

However the claim about the existence of a WIMP species with mass close to 50GeV is discarded by the recent work of a Korean collaboration KIMS. The experiment is carried out using thallium-doped cesium iodide scintillators, and it sets an upper limit on the cross section which is lower than that required for the DAMA $\sim$ 50GeV signal. No explanation for the signal in DAMA is known to this date, however.

A full depiction of the current experimental results is given in Figure 26.1 of reference [2]. In summary, WIMP particles as a model for dark matter has not been experimentally established, however there are upper limits on the cross-sections for various masses that have been established by various experiments all over the world. Future experiments will open a new window into an understanding towards the properties and behavior of dark matter.

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Jet Quenching

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Abstract

We give a brief introduction to the study of quark-gluon plasma and its relation to jet-quenching.

1 Introduction

A thorough understanding of our universe at its earliest stages begs for, amongst many things, a better comprehension of the gluons and quark matter content at high temperatures and densities. Heavy ion collisions open a window into this area of physics by creating a large though unstable fluid of particles where quarks are not confined but are instead free to roam around with gluons. Such experiments give us an insight into what our universe likely looked like a millionth of a second after the Big Bang. This essay aims to introduce the reader to the study of so-called Quark-Gluon Plasma by presenting a brief motivation for the subject (section 2), followed by presenting some experimental evidence for its existence (section 3).

2 A brief introduction to Quark-Gluon Plasma

The theoretical model used to describe the strong force of the nature is Quantum Chromodynamics (QCD). This theory is asymptotically free, i.e. the gauge coupling g takes very small values at very high energies. On the contrary, at lower energy scales, g takes higher values, leading to the confinement of quarks inside hadrons. The asymptotically free quarks and gluons form a state of matter, entirely different from the state of matter that we experience in the form of baryons at lower energies, known as the Quark-Gluon Plasma (QGP).

In the phase diagram of QCD, different phases can be shown easily in a temperature-density diagram. At low temperatures and low density, we see the ordinary matter; while QGP lies at the high temperature-high density region of the diagram [1]. This is shown in Figure 1.

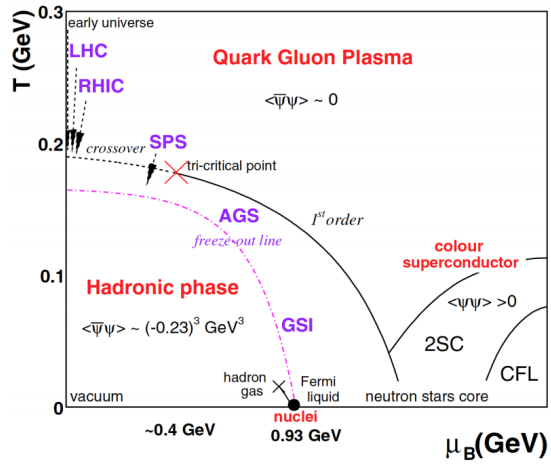

Figure 1: Phase Diagram of QCD; on the axis are temperature T and baryon chemical potential μ_B. [1]

3 Jet Quenching - An experimental test for the existence of QGP

The experimental evidence for the existence of QGP comes from the process of high-energy nucleus-nucleus collisions. As the particles collide, they form a hot and dense medium, which cools down at a very short time scale, relaxing into a hadronic state. Jets of hadrons are created in the initial phase after the collision and act as probes of the QGP. These jets lose energy as they pass through the plasma due to gluon bremsstrahlung in a phenomenon known as "jet quenching" [2]. In order to quantify this loss in energy, we define the nuclear modification factor,

$$R_{AB}(p_T, \eta) = \frac{(1/N_{evt})d^2N_{AB}(p_T)/dqp_T d\eta}{(\langle N_{\text{bindary}}/\sigma_{\text{inel}}\rangle dN^2_{pp}(p_T)/dp_T d\eta} \tag{3.1}$$

Here N_{AB} denotes the number of collisions of the two nuclei A and B, p_T denotes the transverse momentum of a jet and $\langle N_{\text{bindary}}\rangle$ denotes the average number of nucleon collisions per event. If $R_{AB} = 1$, the scattering of two nuclei may be well-thought of as a superposition of the scattering of individual partons. It has been prosposed theoretically that the value of R_{AB} is affected more in the presence of hot, dense matter as compared to the usual baryonic matter [3]. This gives a way to test the existence of QGP.

Measurements of of R_{AB} have been made at Brookhaven National Laboratory's Relativistic Heavy Ion Collision (RHIC) and at the Large Hadron Collide (LHC) at CERN. Here we review the results announced by the RHIC collaboration only [2]. This experiment involves scattering of gold nuclei, i.e. $A = B =$ Au at the center of mass energy $\sqrt{s_{AB}} = 130$ GeV [2] . The results are shown in figure 2, where the numerator and the denominator in the definition (3.1) are plotted as functions of p_T for both hadrons and neutral pions. The error bars are also shown in both halves of the figure. The suppression of high-p_T hadron production gives an evidence for the existence of QGP.

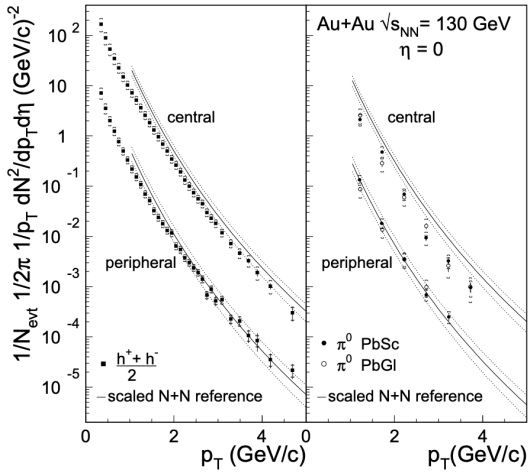

Figure 2: Spectrum from nuclear collision vs spectrum from nucleon collision only [2].

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Sterile Neutrinos

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Abstract

We discuss the phenomenology of sterile neutrinos and its relation to the physics of neutrino oscillations.

1 Introduction

Neutrino physics is a strong candidate for opening a window into the physics beyond the Standard Model. Since these particles are extremely light and mostly elusive, a possibility that there exist new species which interact only through gravitational force called 'sterile neutrinos' is under investigation. These neutrinos are strong candidates for the explanation of anomalies detected in the short-baseline experiments at LSND and at MiniBooNe. They also appear naturally in the extensions of Standard Model when neutrino oscillations are studied. We present a review of these topics in this essay. We will cover some theoretical background in section 2, followed by a discussion of the experimental results in section 3. Most of our presentation follows the recent review [1].

2 Theoretical background

The Standard Model is a chiral gauge theory, which means that the left-handed and the right-handed fermions couple differently to the gauge degrees of freedom. In particular, while the left-handed leptons along with the left-handed neutrinos form a doublet under the $SU(2)$ factor of $SU(2) \times U(1)$ gauge symmetry of the electroweak sector, the right-handed leptons exist as singlets. In other words, there are no right-handed neutrinos in the Standard Model. This fits well with an initial prediction of the Standard Model that the neutrinos are massless particles, since the Dirac mass term would require the existence of the right-handed particles. One could write instead a Majorana mass term for neutrinos consisting entirely of the left-handed particles, however such a term violates the gauge invariance of the action.

The discovery of neutrino oscillations (first made in 1998) is not consistent with this picture. It requires the existence of massive neutrinos and hence the Standard Model as described above had to be extended. One proposed way to do so was to add an arbitrary

number N of right-handed neutrinos that are singlets under the gauge symmetry. This implies that a Majorana mass term can be written for these neutrinos, and that these masses are not generated by Higgs mechanism.

In the presence of these neutrinos, we distinguish between the flavor basis and the mass basis where the two are related by a unitary $(3+N)\times(3+N)$ matrix U, leading to the following coupling between W-bosons and neutrinos.

$$\mathcal{L} = -\frac{g}{\sqrt{2}} \sum_{\alpha=e,\mu,\tau} \sum_{k=1}^{N} (\bar{l_{\alpha L}} U_{\alpha k} \nu_{kL} W_{\rho}^{\dagger} + \text{h.c.}) \tag{2.1}$$

Then the probability of a neutrino oscillation $\nu_\alpha \to \nu_\beta$ can be calculated theoretically. It has the following formula,

$$\begin{aligned} P_{\nu_\alpha \to \nu_\beta} &= \delta_{\alpha\beta} - 4 \sum_{k>j} \text{Re}[U^{\star}_{\alpha k} U_{\beta k} U_{\alpha j} U^{\star}_{\beta j} \sin^2 \left(\frac{\Delta m^2_{kj} L}{4E} \right) \\ &\quad + 2 \sum_{k>j} \text{Im}[U^{\star}_{\alpha k} U_{\beta k} U_{\alpha j} U^{\star}_{\beta j} \sin \left(\frac{\Delta m^2_{kj} L}{2E} \right) \end{aligned} \tag{2.2}$$

where $\Delta m^2_{jk} \equiv m^2_j - m^2_k$ are the differences between squares of neutrino masses, L is the distance between the source and the detector and E is the energy of neutrinos.

3 Experimental results

We discuss, in this section, two experiments where the possibility of the existence of sterile neutrinos has been studied in past three decades. First, the Liquid Scintillator Neutrino Detector (LSND) experiment at the Los Alamos Meson Physics Facility (LAMPF) conducted a fixed-target experiment where a beam of protons with energy 798 MeV scattered against a large number of pions. The neutrinos were produced from π^+ particles as these pions decayed into muons ($\pi^+ \to \mu^+ + \nu_\mu$) which further decayed into electron neutrinos and muon anti-neutrinos ($\mu^+ \to \bar{\nu}_\mu + \nu_e + e^+$). These events were detected by an LSND placed 30m away from the target

As neutrino oscillations occur, a significant number of $\bar{\nu}_e$ particles are detected which are produced from the $\bar{\nu}_\mu \to \bar{\nu}_e$ decay. The average transition probability for this decay was measured to be $(2.64 \pm 0.67 \pm 0.45) \times 10^{-3}$ with a significance level of 3.8σ. This probability amplitude is reproduced through equation (2.2) if the existence of a right-handed sterile neutrino is assumed, with $\Delta m^2_{41} \sim 0.1\text{eV}^2$.

In order to confirm the validity of these results, an experiment named MiniBooNe has been underway at FermiLab since 2002. In this experiment, protons of energy 8 GeV collide with a fixed beryllium target and a detector placed at a distance of 541 m from the target is used to make measurements. The L/E values of this experiment range from 0.18 to 2.7 m/(MeV) which includes the range of L/E (0.5 to 1.5 m/(MeV)) covered by the LSND experiment. As the values of L are known for the two experiments, it follows

that in order to make a comparison, the range of relevant energies at the MiniBooNe experiment $E > 360$MeV.

It was reported in earlier papers that at energies above 475 MeV, no evidence for the existence of sterile neutrinos has been found [2]. However, in recent publications, the results for energy scales below 475 MeV have been reported, where an abundance of ν_e-like events has been detected [3]. However, since the energy scales $E < 360$MeV are outside of the range of LSND experiments, the events detected in MiniBooNe are due to an effect different from the LSND anomaly. A definite reason for this low-energy anomaly is not yet known, however it is known that the MiniBooNe experiment does not distinguish between ν_e-like events and photo detections. The study of this distinction will be a subject of investigation at MiniBooNe in the future when a liquid-argon time-projection chamber will be used instead of the current detector that pure mineral oil $(\mathrm{CH})_2$ for scintillation.

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An introduction to the Higgs Boson

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Abstract

We review Higgs mechanism and the Higgs boson in the Standard Model and the recent experiments that have shown strong agreement with Higgs properties as predicted by the Standard Model.

1 Introduction

It had been long known that if there exist any fundamental forces of nature besides the electromagnetic force, the force carriers must be massive enough so that their effects cannot be detected at lower energies. On the other hand, a theoretical formulation of the Quantum theory of electromagnetism as an abelian gauge theory presented with a conundrum: if the massive bosons are gauge fields for a gauge theory, there is no way to write a mass term for them in a gauge invariant way. This problem is resolved by so-called 'Higgs mechanism' which we review in section 2.

In the Standard Model, this mechanism leads to the prediction that there exists a single scalar field, called Higgs field, which interacts with all massive particles of the model. We discuss how certain properties and interactions of this field have been detected in the past few years in section 3. In section 4, we discuss how a future collider using $\mu^- + \mu^+$ scattering could help us learn more about Higgs physics.

2 Higgs Mechanism in the Standard Model

We first discuss the Higgs Mechanism in the electroweak sector of the Standard Model; focusing further only on the bosonic fields. The interactions of Higgs with fermionic fields will be discussed later. The bosonic fields of the electroweak sector are theoretically modeled as an $SU(2) \times U(1)$ gauge theory with a complex scalar Φ in the doublet of $SU(2)$ and $U(1)$ charge $\frac{1}{2}$. The Lagrangian of this theory contains a potential $V(\Phi)$ in addition to the kinetic terms for the fields, which is responsible for the spontaneous breaking of the gauge symmetry. In particular, we take $V(\Phi) = \lambda \left(\Phi^\dagger \Phi + \frac{m^2}{2\lambda}\right)^2$ where $m^2 < 0$.

$$\mathcal{L} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi) \tag{2.1}$$

Here $G^a_{\mu\nu}$ and $F_{\mu\nu}$ are field strengths of the $SU(2)$ and $U(1)$ gauge fields respectively, and D_μ is the covariant derivative with respect to $SU(2) \times U(1)$ gauge symmetry. The minima of the potential are at $\Phi^\dagger\Phi = -\frac{m^2}{\lambda}$. Given m^2 and λ, a gauge can be chosen such that

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} \tag{2.2}$$

where $v^2 = -\frac{2m^2}{\lambda}$. Around the vacuum Φ can be expanded as

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{2.3}$$

so that the potential takes the following form in terms of the "Higgs field" $h(x)$:

$$V(x) = \lambda \left(\frac{h^2}{2} + vh \right)^2 = \lambda \left(\frac{h^4}{4} + v^2 h^2 + vh^3 \right) \tag{2.4}$$

The second term on the right hand side is the mass term for the Higgs, with the mass being equal to $\sqrt{\lambda v^2}$. In order to see that Higgs gives mass to the vector bosons, we expand the the kinetic term for Φ.

$$(D_\mu \Phi)^\dagger D^\mu \Phi = \frac{1}{8} v^2 \left(g^2 (W_1^2 + W_2^2) + (-gW_3 + \tilde{g} B_\mu)^2 \right) + \cdots \tag{2.5}$$

where $\cdots$ are terms involving the Higgs field. This equation can be re-written in terms of the components of W^a_μ associated to the raising and lowering operators $\sigma^\pm$, i.e. $W^\pm = \frac{1}{\sqrt{2}}(W_1 \mp iW_2)$ as well as the photon A_μ and the Z-boson which diagonalize the term $(-gW_3 + \tilde{g} B_\mu)^2$,

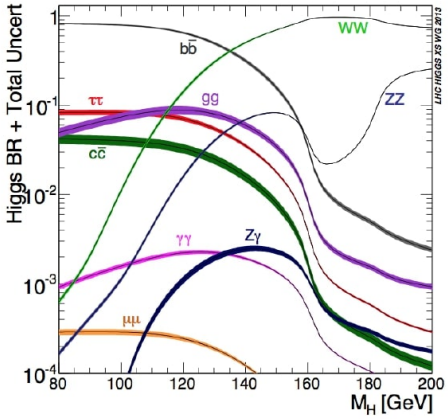
$$(D_\mu \Phi)^\dagger D^\mu \Phi = \frac{1}{8} v^2 \left(g^2 (W_+^2 + W_-^2) + (g^2 + \tilde{g}^2) Z_\mu^2 \right) + \cdots \tag{2.6}$$

Hence the masses of W and Z bosons are related to the Higgs vacuum as $M_{W^+} = M_{W^-} = \frac{1}{2} vg$ and $M_Z = \frac{1}{2} v \sqrt{g^2 + \tilde{g}^2}$.

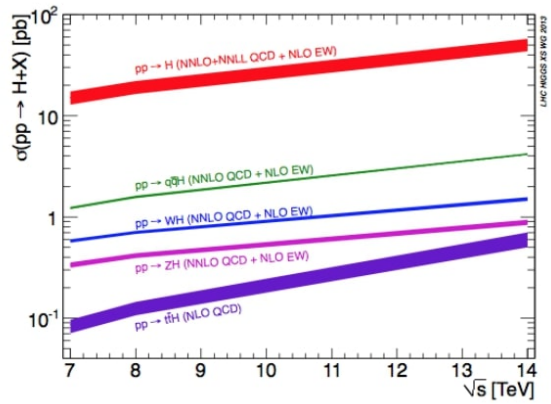
Before we discuss the interactions between massive bosons and the Higgs field, we will discuss the interaction of the fermions of the Standard Model with the complex scalar Φ. In the lepton sector, it is useful to recall that we can write Yukawa interactions in a gauge invariant way. This is because the left-handed leptons transform in the doublet of $SU(2)$ while the right-handed leptons are singlets of $SU(2)$. As Φ takes the vacuum expectation value (2.2), the leptons get a mass

$$\mathcal{L}_{\text{lepton mass}} = - \sum_{\text{lepton flavors}} \frac{\lambda_f}{\sqrt{2}} v f \bar{f} \tag{2.7}$$

where λ_f is the Yukawa coupling constant for the lepton field of flavor f. As discussed in an essay before, the quark mass terms involve slightly more work to understand. In particular, the mass eigenstates and the flavor eigenstates do not coincide, leading to the mixing of quark flavors [2].



(a) Branching ratios of Higgs boson decay as functions of Higgs mass



(b) Cross sections of Higgs boson production as functions of COM energy

Figure 1: Scattering and decay probabilities of Higgs boson as predicted by the Standard Model [1]

Now we discuss the interactions of Higgs field with the vector bosons and the fermions of the Standard Model. In overall, these interactions take the following form,

$$\begin{aligned}\mathcal{L}_{\text{Higgs interactions}} &= -\sum_f m_f \bar{f} f \frac{h(x)}{v} + 2m_W^2 W_\mu^+ W^{-\mu} \frac{h(x)}{v} + m_Z^2 Z_\mu Z^\mu \frac{h(x)}{v} \\ &\quad -3M_H^2 \frac{h(x)}{v} \mathcal{O}(h^4) \end{aligned} \tag{2.8}$$

Based on these interactions, we can predict different decay modes of Higgs field as well as the probabilities of its production in the $p + p$ experiments. These topics are the subject of next section.

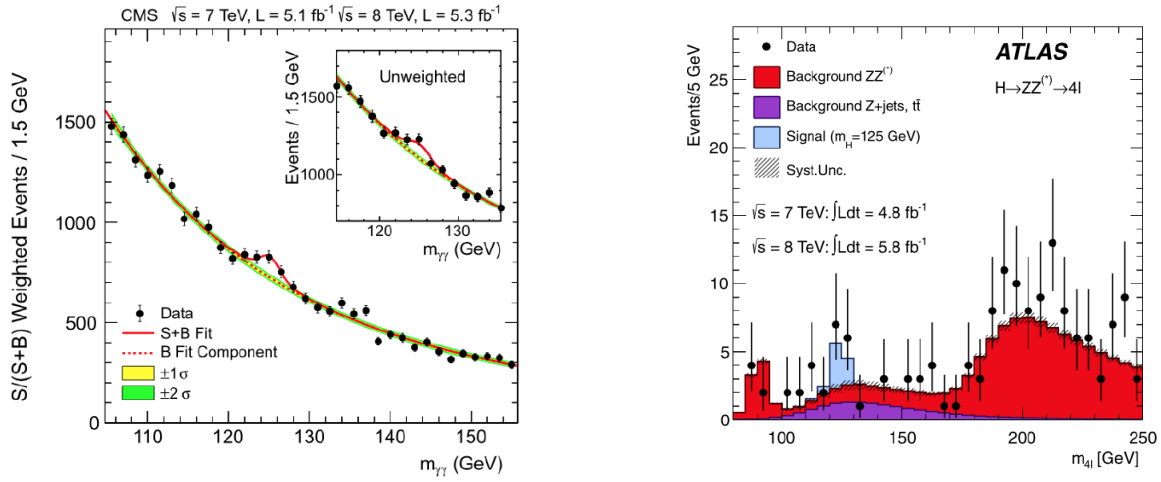
3 Experimental Detection of Higgs Boson

3.1 Higgs interactions and expected detections

It follows from equation (2.8) that Higgs couples most strongly to the top quark and the W and Z bosons. However, the decay channels $h \to W^+W^-$, $h \to Z + Z$ or $h \to t + \bar{t}$ have not been observed at LHC, giving an upper bound on the mass of the Higgs boson. The next heaviest particle is the bottom quark, and gives a possible channel in which Higgs decay could be detected directly. However a computation of the decay amplitude gives

$$\Gamma(h \to b + \bar{b}) = 3\frac{\alpha_W}{8} m_h \frac{m_b^2}{m_W^2} \left(1 - \frac{4m_b^2}{m_h^2}\right)^{3/2} \tag{3.1}$$

At the Higgs mass of 125 GeV, this amplitude comes out to be $\approx$ 2MeV which is a thousand times smaller compared to the width of Z-boson (2.5 GeV). This makes it difficult to measure the width.



(a) Mass distribution of $\gamma + \gamma$ measured by the CMS experiment

(b) Mass distribution in 4 lepton events measured by the ATLAS experiment

Figure 2: Detection of Higgs Boson at CMS and ATLAS experiments in 2012

At this point, we note that while the decay of Higgs boson into W or Z bosons is not expected on-shell because of its small mass, an off-shell decay into $W^+ + +W^-$ or $Z + Z$ is entirely possible. In order to distinguish these from on-shell particles, these products are often denoted as $W + W^\star$ and $Z + Z^\star$ respectively. It is also possible for the Higgs to decay into two photons through a one-loop process. There are other possibilities for Higgs decay modes, which we do not mention in the text but are shown in figure 1a where their branching ration are drawn as a function of the possible Higgs mass.

Another possibility for Higgs to be involved in the experiments is through its production from the scattering of two proton beams. The cross sections for such processes with various resultant particles is shown in figure 1b.

3.2 Higgs Boson Discovery at ATLAS and CMS

The first detection of the Higgs boson was carried out at ATLAS and CMS detectors back in 2012 [3, 4]. The decay modes on which these experiments focused are as below,

$$\begin{aligned} h &\rightarrow \gamma + \gamma & BR &= 0.23\% \\ h &\rightarrow Z + Z^\star \rightarrow 4l & BR &= 0.016\% \end{aligned}$$

where $4l$ denote a four lepton result. The mass distribution of two photons collected at CMS is shown in figure 2a where a resonance around 125GeV is evident. This is confirmed independently by the detection of a Higgs signal in the mass distribution for 4 leptons shown in figure 2b.

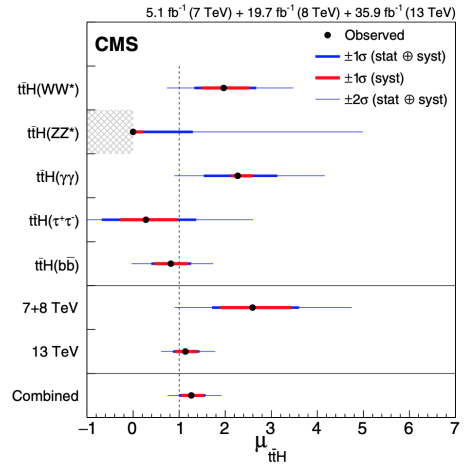


Figure 3: Detection of $p + p \to t + \bar{t} + H$ scattering with various Higgs decay modes.

3.3 $t + \bar{t} + H$ production

As reviewed in section 3.1, the Higgs boson does not decay into a top and anti-top quark because of its light mass. This leads to an important question: is it possible at all to detect directly the coupling of Higgs to top quark experimentally? The answer, as discovered recently, is in affirmative [5]. This work from CMS collaboration has collected evidence of the tree-level coupling between the Higgs boson and the top quark in a proton-proton collision, $p + p \to t + \bar{t} + H$. The Higgs Boson can further decay into any of the following five most probable decay modes, $W + W^\star$, $Z + Z^\star$, $\gamma + \gamma$, $\tau^+ + \tau^-$ and $b + \bar{b}$. The observed values along with 1σ and 2σ confidence intervals for each of these decay modes at 7 TeV, 8 TeV and 13 TeV are shown in figure 3. The vertical line shows the expected value calculated through Standard Model; it lies within the 2σ interval of all decay modes and energy levels.

4 Outlook: Muon Colliders

Finally, we discuss what a future collider using the scattering of muon and anti-muon beams instead of proton beams might help us learn about the nature of the Higgs boson [6]. Such a collider has a clear advantage over the proton-proton collider as the energy carried by the beam is carried by the muons itself, in contrast to protons, where partons carry only a fraction of the total energy. A14 TeV muon collider is thus expected to probe the Physics that a 100 TeV $p + p$ collider may probe. This gives a new window into studying the physics of Higgs, e.g. the question of whether there are any particles not-yet-detected that couple to Higgs, or if there exists a Higgsino - the superpartner to the Higgs boson.

It is also worth noting that such a collider, however, comes with its own drawbacks. First, muons have very short lifetime, i.e. $2.2\mu s$ at rest, and that it is not yet known how to create a large number of muons in bunches with small emittance. Such problems must be subjects of future research programs so as to help advance the muon physics further.

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